Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering *2022 – 2026*
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream) *2019*
St. John's College, Panadura, Sri Lanka
Results: 3 A's (Combined Mathematics, Physics, Chemistry) **District Rank:** 17

Experience

Software Engineer — Trading Systems *Sep 2026 – Present*
CAL | Colombo, Sri Lanka

- Sole engineer on CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus the trading screens and ops tools around it, so the firm owns its platform instead of a vendor-built broker system.
- Took over **CSE certification** part-way and carried it to the final stage: every application-level case of the **188** exchange test cases re-run on CSE's test system with a full exchange-message log; only failover and trade-bust cases that CSE must trigger remain.
- Built an **event-sourced, CQRS** engine: a **Chronicle Queue** write-ahead log is the source of truth on the order path, with **PostgreSQL** as an async read model, so database writes never block execution.
- Implemented **FIX 5.0 SP2 / FIXT 1.1** order routing and drop copy (**QuickFIX/J**), a custom **Java NIO** decoder for the **MITCH** UDP market-data feed, and client-side token-bucket rate limiting.
- Built and own the Certification Runner (matches each test case to live exchange messages and explains every field difference from CSE's expected result), an order-lifecycle view (New, Fill, Cancel, Amend, Reject, Bust), the ops console, Pro dealing ticket and CAL Trader, plus durable message logs and an audit-ready test register.
- Using AI-assisted engineering (Claude), one person covers what is normally a team's work; design and code passed two senior reviews. Next: live CSE demonstration and sign-off, then Phase 2 (zero-allocation parsing with **Agrona**, active-passive standby).

R&D Software Engineer — KōreroCare Platform *Mar 2026 – Aug 2026*
Empathic Computing Laboratory, University of Auckland | Remote / Part-time

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Co-founder & Lead Engineer (Freelance) *Jul 2025 – Aug 2026*
ScaleX — SaaS Development Lab *Remote / Sri Lanka*

- **Stream-Native Tipping Platform:** Architecting a creator-focused SaaS to solve Sri Lankan payout barriers by integrating PayHere's aggregated gateway. Developed an OBS-native alert system using webhook-driven `notify_url` callbacks to trigger real-time on-stream overlays.

- **High-Performance Scheduling SaaS:** Engineering a mobile-first scheduling tool using Next.js 14 and Supabase Realtime for live heatmap synchronization. Implemented a custom best-slot recommendation algorithm to rank optimal meeting times based on participant overlap and duration constraints.
- **Scalable Infrastructure & DevOps:** Designed a $0-to-launch deployment strategy leveraging Vercel serverless functions and Supabase PostgreSQL with connection pooling. Established robust security protocols including Zod-based input validation and rate-limiting to ensure system resilience.
- **Strategic Compliance & Growth:** Navigating CBSL regulatory frameworks to operate as a software platform while delegating fund-handling to licensed gateways to minimize operational risk. Implementing a "freemium" viral loop strategy with branded footers and Stripe-integrated Pro tiers for monetization.
- **AgriNova:** Built a full-stack inventory and commission management platform (**Next.js**, **Prisma**, **Supabase**/**PostgreSQL**) replacing spreadsheet workflows: 5 branches, 329+ SKUs, 194K+ inventory units tracked and 860+ invoices processed (Rs. 64.4M+) with automated commission calculations. Cut repeat dashboard data-fetch latency from ∼384ms to under 1ms with React Query caching and pagination.

Software Engineer Intern (Backend Focus) *Dec 2024 – Jun 2025*

Information Systems Associates (Pvt) Ltd — Aviation Technology Company

- Engineered a distributed, high-throughput unique ID generator for Passenger Name Records (PNRs) using a Kafka-backed pool model, ensuring uniqueness and security across scaled instances.
- Optimized the PNR generation service to achieve 25+ PNRs/sec throughput per node with <100ms latency.
- Architected and implemented a hybrid integration strategy by embedding a gRPC client and Kafka consumer into a legacy Java 8 monolith.
- Investigated and refactored inefficient Kafka Streams processing logic, building a proof-of-concept to validate a more efficient Producer-Consumer model.
- Developed dynamic, database-driven feature toggles to control PNR routing logic.

Keywords: Java 18, Spring Boot, gRPC, Kafka, REST API, Couchbase, Oracle DB, PostgreSQL, Microservices, Distributed Systems.

Open Source Contributions **See all**

- **kgateway #14851** (CNCF) *Merged, Oct 2026*
  Added the **kgateway_build_info** Prometheus metric (version, revision, Go version, OS/arch) so operators can see which build each pod runs; shipped in **kgateway v2.5.0-beta.2**.
- **kubernetes-mcp-server #1482** *Merged, Oct 2026*
  Fixed flaky HTTP server startup in the **Go** test suite with a 5 s readiness poll and cleanup before failing.
- **kubernetes-mcp-server #1549** *Merged, Oct 2026*
  Test that every tool, prompt and resource across all 12 toolsets declares valid **Kubernetes RBAC** metadata; approved on the first review.
- **Kubeflow Spark Operator #3249** *In review (LGTM)*
  20 **Ginkgo** specs covering every executor pod-phase transition: **100%** of `sparkpod_metrics.go`, plus a log message fix.
- **KEDA #8317** (CNCF) *In review*
  New **onlyReadyPods** option for the Kubernetes workload scaler (count only Running and Ready pods); design agreed with a maintainer, docs in keda-docs #1919.
- **kubernetes-mcp-server #1568** *Root-caused, Oct 2026*
  Showed a reported "flaky" test was a real merge-order bug; the author confirmed and applied the suggested fix.

Projects **See more projects on my portfolio**

1. AgriNova Inventory & Commission Platform (2025 | Freelance, ScaleX) **Live demo**

Designed and delivered a production-grade inventory and sales management system for a real-world client, supporting multi-branch stock operations, invoicing, receipts, and commission tracking.

- Engineered transaction-safe stock movement logic and replaced application-layer aggregation with **PostgreSQL** aggregations, reducing execution time from **∼384 ms to ∼0.02 ms**.
- Designed a multi-layer caching strategy with **Next.js** and **React Query**, reducing repeated dashboard latency from **∼703 ms to ∼0.05 ms**.
- Built a clean, dynamic admin-focused UI and validated system reliability through end-to-end testing with **Playwright**.

Keywords: Next.js, PostgreSQL, Prisma, React Query, Tailwind CSS, Playwright

2. Employee Management System (EMS) (2025 | Individual) **Architecture | Github**

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

3. GoTogether (2024 | Group) **Architecture | Live | Github**

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX, Certbot, and DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service using Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

4. Food Delivery Application (2024 | Group) **Github**

Built a scalable, event-driven food delivery platform inspired by systems like Uber Eats, using a modern microservices architecture. As part of the core team, I focused on designing and developing the Restaurant Service.

- Developed the Restaurant Module with secure sign-up/login using **Spring Security** and **JWT** (with extended claims).
- Built CRUD APIs for menu and item management; integrated **Redis caching** to improve performance and reduce database load, with appropriate cache invalidation strategies.
- Enabled order sync via **Kafka**: when an order is placed, an event is published to the 'order_created' topic, which the restaurant service consumes to update order details.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, PostgreSQL, WebSocket, Docker, React.js, Ant Design

5. Serverless Auction Platform (2025 | Individual) **Architecture | Github**

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Architected an event-driven system using **AWS Lambda**, **API Gateway**, and **EventBridge**, with **SQS** and **SNS** for asynchronous communication.
- Designed a high-performance **DynamoDB** data model with GSIs and implemented a custom **Lambda Authorizer** for JWT-based security.
- Utilized **LocalStack** and **Docker** for local cloud emulation and integrated **Middy** middleware for robust error handling.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, SES, JWT, Middy, LocalStack, Docker

6. Go gRPC GraphQL – Multi-Tenant Microservices (2025 | Individual)

Architected a **polyglot, multi-tenant microservices** platform with a focus on scalability, fault isolation, and unified data access.

- Developed backend services in **Go** using **gRPC** and **Protobuf** for efficient inter-service communication.
- Implemented a **GraphQL API Gateway** to provide a single, unified endpoint for client applications.
- Designed multi-tenant data storage with **PostgreSQL**, **Elasticsearch**, and optional **Cassandra** for high-volume workloads.

Keywords: Go, gRPC, GraphQL, Protobuf, PostgreSQL, Elasticsearch, Cassandra, Docker Compose

7. Fintech & Crypto Trading Automation Lab (Ongoing | Group)

Industry-Mentored Initiative (Ongoing)

- Selected for an exclusive initiative to build intelligent automation systems for fintech and crypto markets under industrial mentorship.
- Developing production-ready trading tools and gaining deep exposure to technical analysis (TA) and algorithmic trading strategies.
- Focusing on high-performance software engineering with strict deadlines and client-side expectations.

Keywords: Fintech, Crypto Trading, Forex, Algorithmic Trading, Technical Analysis, Intelligent Automation, Industrial Mentorship, Personal Finance & Wealth Management

8. Improving Blockchain Scalability: Throughput Enhancement in ZK-Rollups

Research Project — University of Moratuwa

- Built a modular ZK-rollup benchmarking framework to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, proving, data availability, and L1 settlement.
- Implemented the Rust-based Submitter service for finalized batch submission, DA-mode selection, L1 bridge interaction, transaction receipt validation, and settlement metric export.
- Added calldata, EIP-4844-style blob, and off-chain / Validium-style benchmark modes, observing approximately 70% lower DA publication cost for blob-style mode compared with calldata.
- Introduced a mock-testable `BridgeClient` abstraction, improving Submitter testability and increasing verifiable test coverage from approximately 41% to 88%.
- Built Docker-based benchmark orchestration scripts, a Poisson-style workload generator supporting 8 TPS base and 80 TPS burst workloads, and Python data tools for throughput, P50/P95/P99 latency, finalization time, gas/cost, and DA-footprint analysis.
- Produced reproducible benchmark outputs including latency CDFs, cost heatmaps, and Pareto-style trade-off comparisons for the final research evaluation.

Technologies: Rust, Solidity, Hardhat, TypeScript, Python, Docker, gRPC, SQLite/PostgreSQL, RISC0 / Mock Proving Path, Matplotlib

9. Blockchain Supply Chain Management DApp (2025 | Individual) Architecture | Live | Github

Built a full-stack decentralized application (DApp) enabling end-to-end shipment tracking and secure escrow payments on the Ethereum blockchain (Polygon Amoy Testnet).

- Developed a production-grade Solidity smart contract with a strict 3-state lifecycle (PENDING → IN_TRANSIT → DELIVERED) and on-chain auditability.
- Implemented a trustless escrow mechanism that locks funds on creation and releases payments only upon verified delivery.
- Built a responsive Next.js/React frontend with global state management and Ethers.js integration for secure wallet connections.
- Designed a CI/CD pipeline using GitHub Actions to automate contract and frontend deployment, utilizing Hardhat for local simulation and verification.

Keywords: Solidity, Hardhat, Ethers.js, Web3Modal, Next.js, React Context API, Tailwind CSS, Polygon Amoy, Smart Contracts, Escrow Payments, CI/CD

Other Relevant Projects

10. Automated AI Shorts Video Creator (2025 | Individual)

An open-source project to automate the creation of short-form videos using a powerful combination of AI tools and workflow automation.

- Orchestrated a video creation pipeline using **n8n** as the central workflow engine.
- Integrated various AI services for script generation, text-to-speech, and content preparation, including **Together AI**, **OpenRouter**, and **ElevenLabs**.
- Containerized the entire stack using **Docker Compose** for one-click local deployment.

Keywords: n8n, Docker, AI, Workflow Automation, Video Generation

11. RPAL Interpreter – Functional Language Compiler (2024 | Group)

Engineered a complete compiler and interpreter for the RPAL (Right-reference Pedagogic Algorithmic Language), built from scratch to parse and evaluate programs according to official language grammar and semantics.

- Developed a Lexical Analyzer using custom Finite State Automata (FSA) and regex-based token identification.
- Built an Abstract Syntax Tree (AST) and Standard Tree (ST) generator.
- Implemented a CSE Machine to evaluate programs using 13 semantic rules of RPAL.
- Built a robust testing framework with **Pytest** covering over 168 test cases across AST, ST, and full output validation.

Keywords: Python, Pytest, GitHub Actions, Makefile, CLI

12. AI Blog Post Generator (2025 | Individual)

A multi-agent system that uses specialized AI agents to generate professional, well-structured, and well-researched blog posts.

- Built with the **Agno Framework** for its beginner-friendly approach to multi-agent workflows.
- Orchestrated research, content extraction, and writing using specialized agents for each task.

- Designed for provider flexibility, with easy switching between **OpenAI, Gemini, Claude, and Groq** models.

Keywords: Streamlit, Agno Framework, Pydantic

13. Maison Vella – E-Commerce Admin Panel (2024 | Individual) **Architecture | Live | Github**

Comprehensive admin interface for the Maison Vella e-commerce ecosystem, facilitating centralized management of products, orders, and content assets.

- Engineered a responsive **Next.js 14** app with **Server Actions**, **tRPC**, and **ShadCN UI**, implementing **CQRS** and layered architecture.
- Integrated **Sanity** (Headless CMS) via GraphQL and managed data with **Prisma** and **Supabase** in a serverless environment.
- Secured access with **Clerk Auth** and optimized media delivery using **Cloudflare R2** and a global CDN.

Keywords: Next.js 14, Server Actions, tRPC, GraphQL, Prisma, Supabase, Sanity, Serverless, Cloudflare R2, CQRS

14. Content-Aware Meme Generator (2025 | Individual)

An automated system for generating contextually relevant memes from any URL, leveraging LLMs and the Memegen.link API.

- Orchestrated a multi-step meme generation workflow using **LangGraph**.
- Utilized **Llama 3.3** (via Groq API) for content analysis, concept generation, and text creation.

Keywords: Streamlit, LangGraph, Llama 3.3, Groq, Pydantic, Beautiful Soup

15. PaperMind – AI Research Assistant (2025 | Individual)

Developed a lightweight, **RAG-powered** platform for intelligent querying of academic papers.

- Designed a **LangChain** pipeline with **ChromaDB** vector storage for fast, context-aware lookups.
- Generated embeddings using **Sentence Transformers** for improved retrieval accuracy.
- Delivered LLM-generated responses using **Gemini API** and HuggingFace models.

Keywords: Streamlit, LangChain, ChromaDB, Sentence Transformers, Gemini API, HuggingFace

16. Virtual Me – 3D Animated AI Chatbot (2024 | Individual)

Created an interactive 3D virtual assistant with lifelike animations and AI-driven conversations.

- Integrated **Hugging Face** (Mixtral model) for natural language dialogue.
- Implemented **Eleven Labs** TTS for human-like voice output.
- Built immersive 3D interface with **React** and **Three.js**.

Keywords: React (Three.js), Node.js, Express.js, Hugging Face, Eleven Labs, Rhubarb, FFmpeg

Training & Certifications

CKA Training — KodeKloud, 2025 • AWS SAA-C03 Training — Udemy, 2025 • Apache Kafka Series — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Advanced OAuth Security — Udemy, 2024

Skills

Core	**Java** | **Spring Boot** | **Microservices** | Distributed Systems | System Design | **REST APIs**
Backend	Spring Cloud | Spring Security | Resilience4j | Node.js | Express.js | Gin
Messaging & APIs	**Kafka** | RabbitMQ | **gRPC** | **GraphQL** | WebSockets | Protobuf | JWT | OAuth 2.0
Data	**PostgreSQL** | MySQL | Oracle DB | Redis | Couchbase
Cloud & DevOps	AWS | **Docker** | Kubernetes | CI/CD (GitHub Actions, Jenkins)
Testing	JMeter | Gatling | Pytest | Unit Testing
Tools	Git | Gradle | Postman | Jira | Confluence

Referees

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